Universal Automated Chemical Reaction Engine

Thermal–Atomistic Simulation with Bonding Predictions
and SolidWorks-Compatible Engineering Output

VSEPR-SIM Research Platform

VSEPR-SIM Development

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§1 Scope and Purpose

The Universal Automated Chemical Reaction Engine extends VSEPR-SIM from atomistic structure generation into full chemical reaction simulation. It bridges three domains:

(a) **Chemistry** — real stoichiometric reactions with mass balance, thermodynamics, and mechanism tracking.

(b) **Atomistic simulation** — the 6+9 seed-and-bead stepper with Lorentz–Berthelot mixing and environment-responsive coupling.

(c) **Engineering output** — SolidWorks-compatible CFD material data (12–22 element output per species), structural point clouds, and Excel-compatible spreadsheets.

§2 Reaction Archetypes

Three chemically distinct reaction archetypes are implemented:

§2.1 Reaction 1 — Organic Synthesis with Inorganic Reagent

$$\mathrm{C_6H_6} + \mathrm{HNO_3} \xrightarrow{\mathrm{H_2SO_4}} \mathrm{C_6H_5NO_2} + \mathrm{H_2O} \tag{1}$$

Electrophilic aromatic substitution. Benzene (organic substrate) reacts with nitric acid (inorganic reagent) in the presence of sulfuric acid (catalyst) to produce nitrobenzene (organic product) and water (by-product).

§2.2 Reaction 2 — Special Inorganic Precipitation

$$\mathrm{Th(NO_3)_4} + 2\,\mathrm{H_2C_2O_4} + 6\,\mathrm{H_2O} \longrightarrow \mathrm{Th(C_2O_4)_2} \cdot 6\mathrm{H_2O}\downarrow + 4\,\mathrm{HNO_3} \tag{2}$$

Precipitation of thorium oxalate sludge from aqueous solution. The product is a fine, dense precipitate used in actinide separation chemistry.

§2.3 Reaction 3 — Thermal Decomposition (Salt → Gas + Powder)

$$2\,\mathrm{Cu(NO_3)_2} \longrightarrow 2\,\mathrm{CuO}(s) + 4\,\mathrm{NO_2}(g) + \mathrm{O_2}(g) \tag{3}$$

Thermal decomposition of copper(II) nitrate. Produces a black CuO powder residue plus brown NO_2 gas and colourless O_2 gas.

§3 Thermodynamic Framework

All thermodynamic quantities are computed from tabulated standard enthalpies of formation (ΔH_f°) and Gibbs energies (ΔG_f°).

§3.1 Enthalpy of Reaction

$$\Delta H^\circ_{\mathrm{rxn}} = \sum_{\text{products}} \nu_i \, \Delta H^\circ_{f,i} - \sum_{\text{reactants}} \nu_j \, \Delta H^\circ_{f,j} \tag{4}$$

§3.2 Gibbs Energy and Equilibrium

$$\Delta G^\circ_{\mathrm{rxn}} = \sum \nu_i \, \Delta G^\circ_{f,i} - \sum \nu_j \, \Delta G^\circ_{f,j} \tag{5}$$

$$K_{\mathrm{eq}} = \exp\left(\frac{-\Delta G^\circ_{\mathrm{rxn}}}{RT}\right) \tag{6}$$

§3.3 Entropy of Reaction

$$\Delta S^\circ_{\mathrm{rxn}} = \frac{\Delta H^\circ_{\mathrm{rxn}} - \Delta G^\circ_{\mathrm{rxn}}}{T} \tag{7}$$

§3.4 Arrhenius Kinetics

$$k(T) = A \exp\left(\frac{-E_a}{RT}\right) \tag{8}$$

§4 Species Data Model

Each `ChemicalSpecies` carries:

Table 1: Species data fields

Field	Units	Description
`name`	—	Human-readable name
`formula`	—	Molecular formula
`phase`	—	Solid / Liquid / Gas / Aqueous / Sludge
`molecular_weight`	g/mol	Sum of atomic masses
`density`	g/cm^3	Bulk density
`delta_Hf`	kcal/mol	Standard enthalpy of formation
`delta_Gf`	kcal/mol	Standard Gibbs energy
`S_std`	cal/(mol·K)	Standard entropy
`Cp`	cal/(mol·K)	Heat capacity
`lj_sigma`	Å	LJ σ parameter
`lj_epsilon`	kcal/mol	LJ ϵ parameter
`thermal_cond`	W/(m·K)	Thermal conductivity
`specific_heat`	J/(g·K)	Specific heat capacity
`viscosity`	Pa·s	Dynamic viscosity
`dielectric`	—	Relative permittivity
`dipole_moment`	Debye	Electric dipole moment

§5 Bonding Prediction

Bonding predictions combine:

1. **Electronegativity difference** — Pauling's rule: $\Delta\text{EN} > 1.7 \Rightarrow$ ionic character.
2. **Bond change tracking** — explicit record of every bond broken and formed, with energies.
3. **Environment-responsive state** — η at the time of bond formation/breaking provides a structural context measure.

§6 Mass Balance

Every reaction is verified for mass conservation:

$$\sum_{\text{reactants}} \nu_j \cdot n_{j,\text{elem}} = \sum_{\text{products}} \nu_i \cdot n_{i,\text{elem}} \quad \forall \text{ elements} \tag{9}$$

Tolerance: $|\Delta| < 10^{-6}$ moles per element. By-products are tracked explicitly and included in the balance.

§7 SolidWorks Engineering Output

§7.1 CFD Material Data (22-Element CSV)

Each species produces a 22-field record suitable for SolidWorks Flow Simulation import:

Table 2: 22-element material property output

#	Field	Units	Category
1	Species name	—	Identity
2	Formula	—	Identity
3	Phase	—	Identity
4	Density	g/cm^3	Mechanical
5	Molecular weight	g/mol	Mechanical
6	Viscosity	Pa·s	Mechanical
7	Surface tension	N/m	Mechanical
8	Temperature	K	Thermal
9	Thermal cond.	W/(m·K)	Thermal
10	Specific heat	J/(g·K)	Thermal
11	Cp	J/(mol·K)	Thermal
12	Melting point	K	Thermal
13	Boiling point	K	Thermal
14	ΔH_f	kcal/mol	Thermodynamic
15	ΔG_f	kcal/mol	Thermodynamic
16	S°	cal/(mol·K)	Thermodynamic
17	ΔH_{rxn}	kcal/mol	Thermodynamic
18	ε_r	—	Electrical
19	μ (dipole)	Debye	Electrical
20	σ_{LJ}	Å	Atomistic
21	ε_{LJ}	kcal/mol	Atomistic
22	$\langle\text{EN}\rangle$	Pauling	Atomistic

§7.2 Material XML (.sldmat)

An XML material library file compatible with SolidWorks material import:

Listing 1: SolidWorks material XML format

```
<material name="Benzene">
  <physicalproperties>
    <DENS value="876.5" units="kg/m^3"/>
    <KX value="0.159" units="W/(m*K)"/>
    <C value="1740" units="J/(kg*K)"/>
    <MU value="0.000604" units="Pa*s"/>
  </physicalproperties>
</material>
```

§8 Implementation Summary

Table 3: Source file inventory

File	Purpose	Lines
`chemistry/species.hpp`	Species data model	∼210
`chemistry/reaction.hpp`	Reaction framework	∼260
`chemistry/reaction_library.hpp`	3 built-in reactions	∼490
`chemistry/reaction_engine.hpp`	Simulation + export	∼440
`example_universal_reaction_engine.cpp`	Main demo	∼190
`section_universal_reaction_engine.tex`	This document	∼280

§9 Dependencies

Table 4: Module dependencies

Module	Depends On
`species.hpp`	`atomistic/core/state.hpp`
`reaction.hpp`	`species.hpp`
`reaction_library.hpp`	`reaction.hpp`
`reaction_engine.hpp`	`reaction_library.hpp`, `seed_bead_stepper.hpp`, `snapshot_graph.hpp`, `excel_export.hpp`, `solidworks_export.hpp`